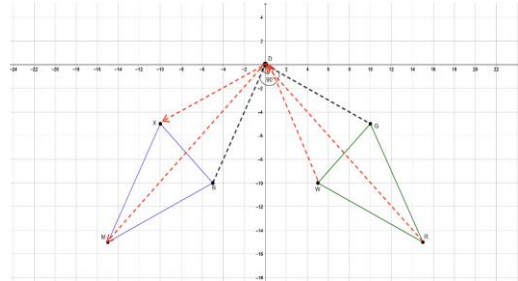


4.1 Algebraic Descriptions for Translations and Rotations

Lesson Introduction

 Benchmark	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.		 Learning Targets	- I can relate a translation on the coordinate plane to an algebraic notation of the transformation. - I can relate a rotation about the origin on the coordinate plane to an algebraic notation of the transformation.	
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A		
 Essential Questions	- How can translations be represented algebraically? - What is the same or different about horizontal and vertical translations? - How can rotations be represented algebraically? - How are rotating clockwise and counterclockwise similar? How are they different?				
 Lesson Narrative	Students collaborate while making connections using multiple representations of translations and rotations. Students make use of tables to represent values as coordinate pairs to understand how to write translations and rotations using algebraic notation. The majority of this lesson is collaborative in nature, with discussions serving as the primary driver of learning throughout the 4.1.2 and 4.1.3 Collaboration components that act as the bulk of the lesson.				
 Component Summary	4.1.1 Warm-Up (~5 min.)	Students learn about pharmacists as a Career Profile (video) (<i>SE p. 177</i>)	4.1.3 Collaboration (15 min.)	Students explore the relationship between image rotations and algebraic representation or notation (<i>SE p. 180</i>)	
	4.1.2 Collaboration (15 min.)	Students explore the relationship between image translations and algebraic representation or notation (<i>SE p. 178</i>)	4.1.4 Bring It Together (10 min.)	Students summarize and synthesize learning targets, including abstract algebraic notation (<i>SE p. 183</i>)	
	4.1.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 183</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Translation - Rotation <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Markers		4.1.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions • Students may confuse clockwise and counterclockwise. • Students may confuse which images are considered preimage and what “prime notation” means in context. • Students might confuse x- and y-values in the coordinate plane.	Things to Consider • The connections between the images, the tables, and the algebraic descriptions are important in developing an understanding of transformations. Allow the students to grapple with writing the conjectures so those connections are made. • Consider common misconceptions from Unit 1 and student understanding shown in the unit to help determine collaboration group size and makeup.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 4.1.2 Collaboration positions students to notice the effects of rotation on vertex coordinates. Consider having students apply Notice and Wonder routines within their collaboration group to help each member of the group discover the relationship between changing values of coordinates and the algebraic representations of rotations.	MA.K12.MTR.2.1: Multiple Representations Students will be working to represent translations and rotations algebraically. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight the process by which students are moving from image to abstract using algebraic notation.
 Differentiate Instruction	For a kinesthetic entry point, consider using physical manipulatives such as patty paper or cutouts for students to explore each of the transformations.	
	For example, students can use patty paper for 4.1.3 Collaboration as they map the transformation for the triangle GWR .	
		
Lesson Notes:		

4.1.1 Warm-Up: A Career in Pharmacy

Component Overview: Display the “Career Profile – Pharmacy” video for the entire class. After students watch this two-minute video, prompt the students to read and answer the questions in the Warm-Up independently.



4.1.1 Warm-Up: A Career in Pharmacy

A career as a pharmacist is just one of many jobs in the health care field. Pharmacists need an understanding of both science and mathematics. Pharmacists also need to work with patients on their prescriptions to ensure the medication is working as it should.



1. What is one thing you learned about pharmacists that you did not know?

Answers will vary.

2. How does paying attention to detail help you to learn mathematics?

Answers will vary. Sample Answer: Paying attention to details when working a problem will ensure that no steps are missed, and each step is done correctly.



Consider positioning students to discuss question prompts rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as:

- “I had to pay attention when problem-solving when...”
- “One detail that I have to remember a lot in math class is...”



Do you know someone who is a pharmacist or works in pharmaceuticals? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

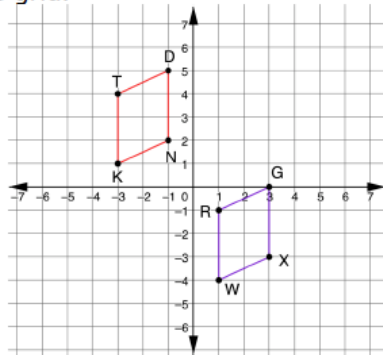
4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations

Component Overview: Students will be collaborating on a series of preimages and images. The exploration stage is the same for each set of images, with students first making observations before representing the transformations algebraically. The first set of images explores the process in-depth with collaborative discussion prompts, while the remaining sets of shapes provide an exploration.



4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations

1. Parallelograms $DNKT$ and GXR are shown on the coordinate grid.



- Describe the transformation if parallelogram $DNKT$ is the preimage.
DNKT is shifted to the right 4 then shifted down 5 OR DNKT is shifted down 5 then shifted to the right 4.
- Describe the transformation if parallelogram GXR is the preimage.
GXR is shifted to the left 4 then shifted up 5 OR GXR is shifted up 5 then shifted to the left 4.
- Complete the table by writing the x -value of each vertex in the columns provided. Then, determine the number of units that should be added or subtracted to the x -value for the vertices of parallelogram $DNKT$ to get the x -value for the vertices of parallelogram GXR .

Vertex	x	Vertex	x	Description
D	-1	G	3	Add 4
N	-1	X	3	Add 4
K	-3	W	1	Add 4
T	-3	R	1	Add 4



Consider highlighting student responses that differ by which parallelogram is identified as the preimage. This discourse can position students to better understand why prime notation is important in mapping translations.



Consider prompting questions for students working in pairs who are stuck.
How are the answers to questions 1a and 1b the same?
How are they different?
What does that make you think about describing transformations?
Why does it matter which parallelogram is the preimage?



As students work in pairs, consider assigning students to image and preimage or x -values and y -values. Charge students to compare and contrast their findings with their partner and make conjectures to answer question 1e based on those findings.

4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations (continued)

- d. Complete the table by writing the y -value of each vertex in the columns provided.

Then, determine the number of units that should be added or subtracted to the y -value for the vertices of parallelogram $DNKT$ to get the y -value for the vertices of parallelogram $GXWR$.

Vertex	y	Vertex	y	Description
D	5	G	0	Subtract 5
N	2	X	-3	Subtract 5
K	1	W	-4	Subtract 5
T	4	R	-1	Subtract 5

Transformations can be described using words, as well as using a notation.

- e. Use the tables to complete the algebraic description of the transformation if parallelogram $DNKT$ is the preimage.

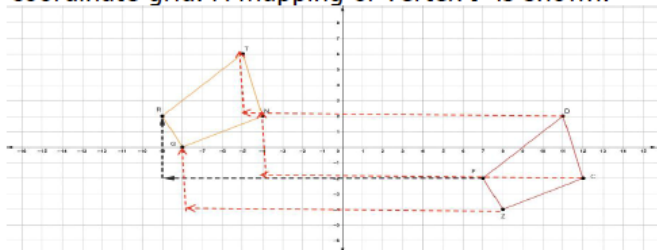
$$(x, y) \rightarrow (x + 4, y - 5)$$

- f. Discuss with your partner how the algebraic description of the transformation would change if parallelogram $GXWR$ is the preimage.

Sample response: The opposite movement is being made and that creates a situation that uses opposite operations.

$$(x, y) \rightarrow (x - 4, y + 5)$$

2. Quadrilaterals $GRTN$ and $ZFDC$ are shown on the coordinate grid. A mapping of vertex F is shown.



- a. Map the transformation for the other vertices of quadrilateral $ZFDC$.



Use prompting questions for students who are struggling to make the connection between the translations and the changes in values.

How does the description of the operation for the x -value relate to the translation?

How does the description of the operation for the y -value relate to the translation?



Prompt students to make connections from question 1f to question 1a and 1b. Students who may not have understood the importance of the order may understand that the algebraic description is incorrect if the preimage and image are switched.

4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations (continued)

- b. Discuss with your partner how the algebraic description of the transformation, $(x, y) \rightarrow (x - 16, y + 4)$, is revealed in the mapping. Write a summary of your discussion.
Answers will vary. Sample answer: ZFDC is shifted to the left 16 units then shifted up 4 units.

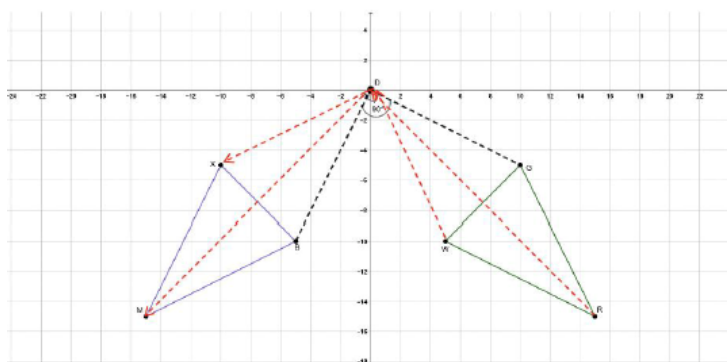
4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations

Component Overview: Students will be collaborating on a series of preimages and images. The exploration stage is the same for each set of images, with students first making observations before representing these movements algebraically. The first set of images explores the process in-depth with collaborative discussion prompts, while the remaining sets of shapes provide an exploration.



4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations

1. Triangles BXM and GWR are shown on the coordinate grid, where $\triangle GWR$ is the preimage. A mapping of vertex G is shown.



- a. Map the transformation for the other vertices of triangle GWR .
- b. Describe the transformation if triangle GWR is the preimage.
 GWR rotated 90° clockwise about point D
- c. Discuss with your partner how to use the table of values to describe the relationship of the corresponding vertices using an algebraic description.

Triangle GWR

Vertex	x	y
G	10	-5
W	5	-10
R	15	-15

Triangle BXM

Vertex	x	y
B	-5	-10
X	-10	-5
M	-15	-15



Consider providing resources like patty paper or a protractor for students to identify the degree of rotation as 90° as well as mapping the vertices to visually represent the right angle (as seen in the answer key).



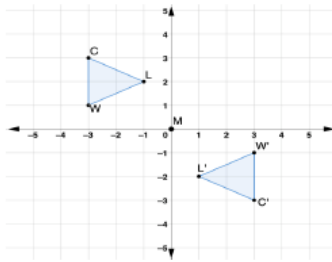
Consider revisiting question 1b, highlighting the direction of the rotation (clockwise), when students are answering question 3b later in the activity.

4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations (continued)

- d. Write a conjecture that you and your partner think will work for the algebraic description of the rotation.
 $(x, y) \rightarrow (y, -x)$

Answers will vary. Sample Answer: Change the sign of the x -value by multiplying by -1 . Then, swap the x - and y -values.

2. Triangles CLW and $C'L'W'$ are shown on the coordinate grid. Triangle CLW was rotated 180° about M to create $\triangle C'L'W'$.



Triangle CLW

Vertex	x	y
C	-3	3
L	-1	2
W	-3	1

Triangle $C'L'W'$

Vertex	x	y
C'	3	-3
L'	1	-2
W'	3	-1

- a. It was stated that $\triangle CLW$ was rotated 180° about M to create $\triangle C'L'W'$. Was that rotation clockwise or counterclockwise?

Answers will vary. It could be either clockwise or counterclockwise.



Notice and Wonder

Provide students with question stems using the Notice and Wonder routine to provide intentional scaffolding for making connections between the vertices, rotations, and algebraic descriptions.

Notice: What do you notice about the values in Vertex G compared to Vertex B ?

Wonder: Does this also hold true for Vertex W and Vertex X ? Vertex R and Vertex M ?

Question 1d: What is your conjecture for how to represent this algebraically?

This line of question stems can apply to questions 2 and 3 as well as for when students collaborate and explore in pairs.



What do you notice about a clockwise rotation of 180° and a counterclockwise rotation of 180° ?

What other transformations could be used to produce this same image from this preimage? (Example: a reflection of the y -axis followed by a reflection over the x -axis.)

4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations (continued)

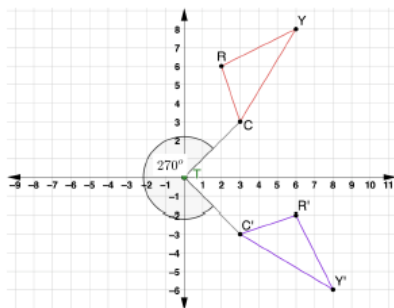
- b. Discuss with your partner how your previous conjecture works with rotating $\triangle CLW$ 180° about M . If necessary, write a new conjecture that works for rotations of 180° about the origin. Include the direction of the rotation determined. Write a summary of your discussion.

Answers will vary. Sample Answer: A rotation of 180° is the same as two rotations of 90° . For a 90° clockwise rotation, you multiply the x -value by -1 then swap the x - and y -values. This is represented by the algebraic description is $(y, -x)$. Rotating 90° clockwise again you will multiply the new x -value, which is y , by -1 and swap the new x - and y -values again. This is represented by the algebraic description of $(-x, -y)$. Therefore the algebraic description for a 180° clockwise rotation will produce an algebraic description of $(-x, -y)$.

- c. Assuming $\triangle C'L'W'$ was the preimage and is rotated 180° about the origin, test whether your previous or new conjecture works to describe this rotation. Include the direction of rotation.

Answers will vary. Sample Answer: Yes, the conjecture works moving in either direction, clockwise or counterclockwise because of the 180° movement.

3. Triangles YCR and $Y'C'R'$ are shown on the coordinate grid. $\triangle YCR$ was rotated 270° counterclockwise about T to create $\triangle Y'C'R'$.



For additional complexity, have students consider rotations around a point other than the origin.



For questions 3-5, consider alternative ways for expressing terms like “clockwise” and “counterclockwise.” Students growing up with digital clocks may be unfamiliar with those terms.



Monitor for students who may be struggling with summarizing their discussion. The student is not expected to express the relationship using algebraic descriptions yet. If students can visually relate the transformation, then build from the visual to the algebraic. If students can relate the transformation algebraically but cannot see it visually, then help them notice the angle through the mapping and use of a protractor.



If a student chose counterclockwise for Part a, argument here still applies. The only difference is that rotating point $(-y, x)$ counterclockwise 90° twice still produces description $(-y, -x)$.

4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations (continued)

Triangle YCR

Vertex	x	y
Y	6	8
C	3	3
R	2	6

Triangle $Y'C'R'$

Vertex	x	y
Y'	8	-6
C'	3	-3
R'	6	-2



- a. Discuss with your partner how the ordered pairs for $\triangle YCR$ and $\triangle Y'C'R'$ could be used to write a conjecture that works for rotations of 270° counterclockwise about the origin. Write a summary of your discussion.

$$(x, y) \rightarrow (y, -x)$$

Answers will vary. Sample answer: the x -value is multiplied by -1 , then the x - and y - value switch.

- b. Test your conjecture for rotations of 270° counterclockwise by looking at $\triangle BXM$ and $\triangle GWR$ from question 1. Assume $\triangle GWR$ is the preimage and was rotated 270° counterclockwise about the origin. If necessary, adjust your conjecture.

- c. Discuss with your partner why the same conjecture should work for a rotation of 270° counterclockwise and 90° clockwise about the origin.

Answers will vary. Sample answer: both rotations result in the same location on the coordinate plane.

If students struggle responding to this prompt, have them compare this transformation to the first rotation. How are they the same? How are they different?

Question 3b intentionally does not give direction so that students include direction in their verification and discussion of their conjecture.

By question 3c, students should be able to draw the conclusion that 270° clockwise is an equivalent rotation to 90° counterclockwise and that 270° counterclockwise is an equivalent rotation to 90° clockwise.

4.1.4 Bring It Together: Algebraic Descriptions for Translations and Rotations

Component Overview: Students should synthesize their findings from the 4.1.2 and 4.1.3 Collaborations by summarizing algebraic descriptions for types of translations and rotations.



4.1.4 Bring It Together: Algebraic Descriptions for Translations and Rotations

Translations	
Right	Left
Add to the x -coordinate $(x, y) \rightarrow (x + a, y)$	Subtract from the x -coordinate $(x, y) \rightarrow (x - a, y)$
Up	Down
Add to the y -coordinate $(x, y) \rightarrow (x, y + b)$	Subtract from the y -coordinate $(x, y) \rightarrow (x, y - b)$

Rotations clockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (y, -x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (-y, x)$
Rotations counterclockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (-y, x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (y, -x)$

1. What do you notice about the algebraic descriptions for rotations?

Answers will vary. Rotations of 90° clockwise are equivalent to rotations of 270° counterclockwise. Rotations of 270° clockwise are equivalent to rotations of 90° counterclockwise. A rotation of 180° has the same result no matter the direction of the rotation.



Use this section to facilitate synthesis, not memorization. Consider using this section as notetaking for students during the Collaboration activities.



Consider a foldable, Frayer Model, or type of interactive notebook for this summary. Examples, nonexamples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

4.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



4.1.5 Wrap-Up: Cool Down

1. A student asks you for help with algebraic descriptions for rotations because they are stuck on how to do it. Describe the process using what you learned in this lesson, highlighting any potential misconceptions they may have.

Answers will vary.